

Trust the View That Sees the Target: Mining Cross-View Conflicts for Reliability-Gated Disaster Damage Assessment

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Abstract

Paired ground-level and overhead imagery is now collected at scale after disasters, and fusing the two views for building damage assessment is standard practice. Fusion is almost always *symmetric*: both views are trusted equally everywhere. We mine three paired collections—inspection photographs from the 2025 Eaton wildfire and street-view panoramas from Hurricanes Ian and Milton, each matched to VHR overhead tiles—for *conflict cases*, the 10–33% of samples on which independently trained single-view models disagree. On these cases an oracle that trusts the correct view exceeds every tested fusion method by 0.37–0.41 accuracy, a gap that survives convergence, calibration, and backbone changes. We close part of it with a visibility-conditioned reliability gate: a linear model over building-visibility features from a frozen segmenter, calibrated per-view confidences, and cross-view disagreement. Under a five-seed protocol with pooled paired tests, the gate is the only method that significantly beats calibrated probability averaging (wildfire conflicts $+0.051$, $p = 0.0001$) and end-to-end fusion ($+0.072$ on conflicts, $p < 10^{-4}$), with parity on the panoramic datasets. A controlled field-of-view intervention shows the mechanism is causal: cropping panoramas toward the building doubles the fusion benefit while geometrically identical random crops do not. Finally, the spatial density of mined conflicts predicts tile-level damage without labels (Spearman $r = 0.615$, $p = 0.001$) while single-view uncertainty anti-correlates. Conflict mining turns “does fusion help?” into “which view should be trusted, where, and why?”.

Keywords

cross-view fusion, street-view imagery, satellite imagery, disaster damage assessment, reliability estimation, GeoAI

1 Introduction

Rapid building-level damage assessment increasingly draws on two large geospatial image collections at once. Overhead imagery gives synoptic coverage within hours and anchors standardized damage mapping [2]; ground-level imagery, from crowdsourced street-view panoramas to official inspection photographs, supplies the facade and debris evidence that overhead sensors cannot see [9, 11]. Cross-view methods that pair the two collections serve geolocalization and damage mapping [4, 5, 10], fusion classification [6], and multimodal arbitration [8], and the standard finding is encouraging: fusion improves average accuracy over either view.

We argue that average accuracy hides both the real value and the real limitation of cross-view fusion. Most records in a paired

collection are easy—both views agree—and fusion adds little. The operationally interesting records are the *conflicts*: samples on which independently trained single-view models disagree. Conflicts concentrate where one sensor is blind: a facade intact from above but gutted at street level, or a roof destroyed above vegetation that hides it from the road. Finding these records in a large collection, deciding which view to believe for each, and reading their spatial distribution is a search-and-mining problem as much as a modeling one. We therefore make conflict cases the unit of analysis and ask three questions. **(1) How much information do fusion methods leave unused on conflicts?** We measure an *oracle single-view gap*, the margin between an oracle that trusts whichever view is correct per sample and the best available method; across three disasters it is 0.37–0.41 accuracy, several times any fusion gain. **(2) Can it be recovered, and how?** Per-sample view reliability is largely predictable from whether the ground image actually *sees the target building*; a *linear gate* over segmentation-derived visibility, calibrated confidence, and disagreement features mixes existing predictors per sample and reads as a one-sentence rule. **(3) Is the regime effect causal?** Datasets differ in hazard, geography, sensors, and labels at once, so we manipulate alignment *within* a dataset with a controlled cropping intervention. A fourth result pays off directly for mining post-disaster collections: the spatial density of conflicts is an unsupervised damage map, computable before any labels exist. Code, protocols, and derived tables: <https://github.com/rayford295/CrossViewGate>.

2 Data and Protocol

We use three paired street/overhead collections spanning two hazards and, critically, two ground-view *observation regimes* (Table 1). **Eaton wildfire (Altadena, CA)**: building-level CAL FIRE Damage Inspection (DINS) photographs after the January 2025 fire, paired by geolocation with post-event VHR overhead tiles; the six-level native scale is mapped to an ordinal three-class task (no/trace, damaged—repairable, destroyed). **Hurricane Ian**: the released CVIAN pairing from CVDIsaster [5], Mapillary panoramas over Sanibel Island with 27 cm post-event satellite imagery and three-level damage-perception labels (4,121 pairs as distributed). **Hurricane Milton**: a street-view/satellite pairing over Florida after the October 2024 hurricane, constructed analogously.

Splits and leakage. Eaton is split by property and tile so all photographs of a structure fall in one partition. The headline Ian numbers keep the release-derived split for comparability with CVDIsaster; an audit against the official CVIAN position GeoJSON found that 299/300 of its test samples share a Mapillary sequence with

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Table 1: Datasets. Regime describes how the ground view observes the labeled building; pairs are train/val/test under the headline protocol.

Dataset	Ground view	Regime	Pairs
Eaton wildfire	CAL FIRE DINS photos	property-centric	6.5k / 1,966 / 1,984
Hurricane Ian	Mapillary 360° panoramas	panoramic	3,248 / 573 / 300
Hurricane Milton	360° panoramas	panoramic	1,740 / 307 / 254

training data, so we also report a repaired 0.005° spatial-block protocol with a 25 m buffer (Section 4). The two protocols are never mixed.

3 Method

Base models and conflict cases. For each dataset we train *street_only* and *remote_only* (ResNet-18 [3], linear head), *concat* (both encoders, MLP head), and *crossview* (both encoders, learned interaction head; the end-to-end fusion reference) under one protocol: class-balanced cross-entropy, image size 224, AdamW, up to 15 epochs with patience-4 early stopping on validation macro-F1, five seeds; a ResNet-50/CLIP/DINOv2 sweep confirmed no conclusion depends on the backbone. With $s(x)$, $r(x)$ the single-view predictions, the **conflict subset** is $C = \{x : s(x) \neq r(x)\}$; conflict rates are 9.6% (Eaton), 33.2% (Ian), 25.7% (Milton).

Oracle single-view gap. The oracle counts a conflict sample correct if *either* single view is correct. For a method m we report conflict accuracy $\text{acc}_C(m)$ and the **oracle-gap closure**

$$\text{closure}(m) = \frac{\text{acc}_C(m) - \text{acc}_C(\text{best single})}{\text{acc}_C(\text{oracle}) - \text{acc}_C(\text{best single})}. \quad (1)$$

The oracle uses no information beyond *which* of two fixed models to trust; it bounds arbitration between existing predictors, not new models.

Calibration decomposition. Averaging miscalibrated probabilities can win or lose for reasons unrelated to information content [1], so we fit one temperature per view on validation predictions and re-evaluate every training-free fusion baseline on calibrated probabilities. Scaling leaves the argmax unchanged, so C is identical before and after.

Visibility-conditioned reliability gate. A frozen SegFormer-B0 [7] trained on ADE20K yields four building-visibility features per street image: building pixel ratio, the ratio inside a centered crop of half the image size, their difference (a centering signal), and the normalized centroid distance from the image center. The gate’s eleven features add calibrated confidence and entropy for both views, the confidence gap, the Jensen–Shannon divergence between the calibrated distributions, and a disagreement flag. It outputs softmax weights over street, remote, and crossview probabilities and is fit on the validation split by minimizing the NLL of the gated mixture with z-scored features. The headline gate is *linear*: a small MLP matched it throughout, and linear coefficients are directly readable. It adds one segmentation pass and a linear fit on top of any deployed pair of models and needs only their output probabilities.

Field-of-view intervention. Each 1024×512 panorama is cropped to a 256×256 window (90° horizontal FOV, horizon band) either **building centered**, at the circular-mean azimuth of the SegFormer

building mask, or **random**, at a per-sample reproducible uniform azimuth. The variants share size, projection, and information budget and differ only in whether the window is aimed at the built structure. We retrain *street_only* and *crossview* on each variant (three seeds), reuse the unchanged remote models, and compare conflict gains.

Conflict density. We average seed-wise street and remote probabilities, aggregate to geographic tiles, and compute per tile the hard conflict density (fraction of argmax disagreements) and a single-view uncertainty control (mean predictive entropy), each correlated (Spearman) with the tile’s mean ordinal damage.

Statistics. Instead of “significant in k of N seeds”, we average each sample’s paired correctness difference over the seeds in which it qualifies (for conflict scope, the seeds in which the single views disagree on it) and test $\mathbb{E}[\delta] = 0$ with a two-sided sign-flip permutation test (20,000 permutations) and percentile bootstrap 95% intervals.

4 Results

Overall and the oracle gap. Table 2 shows three patterns. Cross-view fusion beats both single views on conflicts on every dataset, significantly under the pooled tests (crossview – street: Eaton +0.246, $p < 10^{-4}$; Ian +0.078, $p = 0.013$; Milton +0.141, $p = 0.006$). Once single-view models are converged and calibrated, *simple probability averaging is a strong baseline*: on Eaton it exceeds vanilla crossview (0.954 vs. 0.940), and end-to-end fusion does not significantly beat it anywhere (Eaton conflicts +0.034, $p = 0.073$); claims that learned fusion beats simple aggregation may be artifacts of undertrained or miscalibrated baselines, as ours were in a three-epoch pilot. And the oracle gap dwarfs all of this: knowing *which existing model to trust* per sample would add 0.37–0.41 conflict accuracy. Calibration does not explain it—temperatures are moderate (1.0–1.75) and expected calibration error falls from 0.02–0.14 to 0.01–0.06, yet calibrated averaging closes at most 44% of the gap. The unexploited information is about *when* each view is reliable.

The reliability gate. On the property-centric dataset the gate does what the mechanism predicts (Figure 1a): +0.072 conflict accuracy over end-to-end fusion (95% CI [+0.046, +0.097], $p < 10^{-4}$), +0.018 on the full test set ([+0.014, +0.023], $p < 10^{-4}$), and +0.051 over calibrated averaging ([+0.026, +0.077], $p = 0.0001$), for a 52% closure, the only method above one half. On the panoramic datasets, where street views rarely see the target (mean building pixel ratio 0.027 on Ian vs. 0.119 on Eaton), the gate has little visibility signal to exploit and sits at parity (vs. crossview: Ian +0.005, $p = 0.70$; Milton −0.015, $p = 0.57$), never significantly worse. This asymmetry is the finding: gating pays where ground evidence is target-aligned and degrades gracefully where it is not. The coefficients (Figure 1b) are stable in sign across seeds: street entropy carries the largest negative weight; building centering and the confidence gap are positive. The learned rule is the one an analyst would state: *trust the street view when it is confident and actually looking at the building*. A gate fit on Ian transfers zero-shot to Eaton with 0.27 closure; visibility features are regime-scaled, so a universal gate remains open.

Causal field-of-view intervention. Building-centered cropping raises the conflict gain on both hurricanes (Table 3); on Milton the

Table 2: Test-set results (five-seed means; accuracy std ≤ 0.03 everywhere). Conflict accuracy is computed on the per-seed conflict subset; closure follows Eq. (1) with the oracle at 0.961 / 0.896 / 0.964 conflict accuracy (gap 0.410 / 0.367 / 0.407). Gate = three-view linear reliability gate.

Method	Eaton wildfire (conflict rate 9.6%)			Hurricane Ian (33.2%)			Hurricane Milton (25.7%)		
	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure
street_only	0.931	0.486	—	0.703	0.529	—	0.759	0.557	—
remote_only	0.929	0.475	—	0.648	0.367	—	0.721	0.407	—
concat	0.930	0.674	0.29	0.709	0.588	0.15	0.766	0.622	0.16
crossview (end-to-end fusion)	0.940	0.699	0.35	0.735	0.624	0.25	0.776	0.649	0.22
calibrated probability averaging	0.954	0.732	0.44	0.731	0.617	0.24	0.776	0.626	0.17
reliability gate (linear)	0.959	0.768	0.52	0.731	0.618	0.23	0.777	0.631	0.18

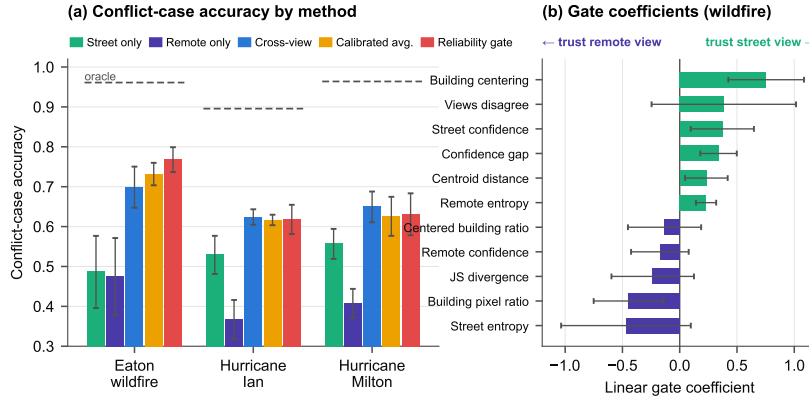


Figure 1: (a) Conflict-case accuracy by method (mean $\pm$ std over five seeds; dashed line = oracle). (b) Linear gate coefficients on the wildfire (mean $\pm$ std over seeds); positive values shift trust toward the street view.

Table 3: Field-of-view intervention (three seeds; remote models unchanged). Conflict gain = crossview – best single view on conflicts.

Dataset	Street-view variant	Street acc.	Conflict gain	Closure
Ian	original panorama	0.709	+0.064	0.166
Ian	building-centered crop	0.648	+0.109	0.269
Ian	random crop (control)	0.574	+0.042	0.072
Milton	original panorama	0.740	+0.067	0.177
Milton	building-centered crop	0.693	+0.149	0.365
Milton	random crop (control)	0.654	+0.055	0.128

closure doubles (0.177 $\rightarrow$ 0.365), reaching two-thirds of the wildfire level, while random cropping with identical geometry lowers it. The building-minus-random contrast is positive in 6/6 dataset-seed pairs. Building-centered cropping *lowers* street-only accuracy (panoramic context is informative for the overall task) yet *raises* the fusion benefit: the gain comes not from a stronger street model but from the two views attending to the same object, and the random control rules out resolution and cropping artifacts. Street-view target alignment is the causal variable behind the regime effect, which grounds the gate’s visibility features as causal rather than merely correlated.

Mining conflicts as a spatial damage signal. On the wildfire, hard conflict density over overhead tiles is a strong label-free predictor of tile damage (Spearman $r = 0.615$, $p = 0.001$ on 25 overhead tiles;

$r = 0.60$, $p < 0.001$ on a 0.01° grid, Figure 2), and it strengthened under converged models (0.545 $\rightarrow$ 0.615). The uncertainty control fails instructively: it *anti-correlates* with damage ($r = -0.37$ to -0.52) because burned-to-the-ground parcels are easy, confident classifications (Figure 2). Conflict density is therefore not repackaged model uncertainty but the spatial signature of two sensors disagreeing about the world, which concentrates where damage disrupts the usual correspondence between facade and roof evidence. Damage on Eaton is spatially autocorrelated (Moran’s $I = 0.27$, $p = 0.0001$), so tile correlations partly reflect spatial structure. On Milton the signal falls below significance under converged models ($r = 0.289$) as better single-view models disagree less; the wildfire case is the robust result.

External comparison and split repair. On the same CVIAN pairing, CVDIsaster’s CGCViT [5] reports 77.96 overall accuracy (street 74.50, satellite 67.07) with a 20M-parameter encoder under a random split; our legacy-protocol ResNet-18 crossview reaches 73.5 ± 1.5 (street 70.3, remote 64.8). Under the repaired spatial-block split with a 25 m buffer, macro-F1 drops by 0.049–0.097 across modes: random or index-based splits overstate spatial generalization, the single-view ranking (street > satellite) is stable, and the size of the cross-view advantage is protocol-dependent.

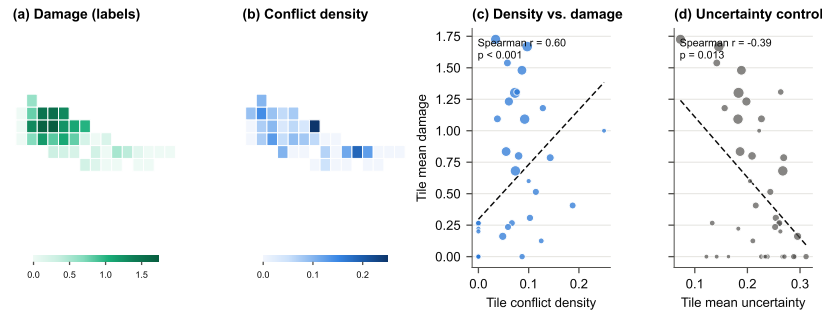


Figure 2: Conflict density as a label-free damage map (Eaton wildfire). (a) Labeled tile damage. (b) Tile conflict density computed without labels. (c) Their rank correlation. (d) The single-view uncertainty control, which anti-correlates with damage. Point size reflects tile sample count.

5 Discussion and Conclusion

The 0.37–0.41 oracle gap measures the opportunity within the restricted choice between two existing single-view models, and it dwarfs what architectural innovation in symmetric fusion has produced. For mining paired geospatial collections the productive question is therefore not “how should features be fused?” but “when should each view be believed?”. The gate closes half of the gap on property-centric imagery with eleven interpretable features; the remaining half is a quantified target for richer visibility descriptors (occlusion, viewing distance, image quality) and pre-event reference imagery. Operationally, the gate costs one frozen segmentation pass per street image plus a temperature and a linear fit on validation predictions, needs only the output probabilities of existing models, and states its rule in one sentence, which matters where automated triage must be auditable. Where ground imagery is panoramic and rarely target-aligned (crowdsourced street view), calibrated averaging captures most of the achievable benefit; where it is property-centric (inspection programs, insurance photography), gating pays. Because the causal driver is alignment rather than hazard, capture policy is itself an intervention: aiming cameras at structures, or cropping panoramas toward detected buildings, raises the value of every downstream fusion component.

Limitations. The property-centric regime is represented by one (large) wildfire dataset; Milton is small (2,301 pairs); the repaired CVIAN test has six spatial blocks connected by shared Mapillary sequences, so strong spatial-generalization claims need more independent groups; the ADE20K segmenter measures built-structure visibility, not *the* target building; the density signal needs replication on further georeferenced disasters; and cross-disaster gate transfer is partial.

Conclusion. Measured on the records where the views disagree, symmetric cross-view fusion leaves 0.37–0.41 accuracy on the table across three real disasters. A linear, interpretable, visibility-conditioned gate recovers a significant fraction of it; a controlled field-of-view intervention shows that the mechanism, street-view target alignment, is causal; and the disagreement signal itself, aggregated spatially, is a label-free damage map that plain uncertainty cannot replicate. Mining conflicts converts “does fusion help?” into “which view should be trusted, where, and why?”.

Acknowledgments

The authors used AI-based assistance for experiment automation, grammar checking, and editorial compression, reviewed all content, and take full responsibility for the work. Supported by the Texas A&M University Environment and Sustainability Initiative (ESI) through the Environment and Sustainability Graduate Fellow Award.

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